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ORIGINAL ARTICLE

Maternal vitamin-D deficiency in Pakistan

MEHNAZ ATIQ[1], AMIN SURIA[2], SHAIKH QAMARUDDIN NIZAMI[1] AND IQBAL AHMED[1]

From the [1]Departments of Pediatrics and [2]Pharmacology, The Aga Khan University Hospital, Karachi, Pakistan

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Objective of the study. This study was performed to assess the vitamin D status of healthy Pakistani nursing mothers and their breastfed infants.
Methods. Seventy-one nursing mothers and their breastfed infants belonging to upper and lower socio-economic class were examined 6 weeks to 11 months after delivery. Serum 25-hydroxy vitamin D [25(OH)D], serum calcium, phosphorus and alkaline phosphatase were measured.
Results. The mean serum 25(OH)D in mothers was 36.7±32.4 nmol/L and 41.25±35.4 nmol/L in infants. Thirty-four (48%) mothers and 37 (52%) infants had levels less than 25 nmol/L. Significantly higher levels were found in uneducated mothers ($p=0.01$), mothers of lower socio-economic class ($p<0.001$) and in those living in mud houses ($p<0.001$). A significant correlation was found between serum 25(OH)D levels of infants under three months of age and their mothers ($p<0.01$).
Conclusions. High prevalence of vitamin D deficiency was found in nursing mothers and their infants predominantly in the upper socioeconomic class.

Key words: nursing mothers; 25 hydroxy cholecalciferol; vitamin D deficiency

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Following several reports on osteomalacia among immigrant Asian women, Heckmatt et al. and other clinicians measured serum concentration of 25-hydroxy vitamin D in pregnant Asian women of Indo-Pakistani origin residing in the United Kingdom and Norway and found a high incidence of severe vitamin D deficiency among them (1, 2, 3). They also reported low cord blood levels of 25-hydroxy vitamin D [25(OH)D], higher incidence of neonatal hypocalcemia, rickets and enamel hypoplasia of primary dentition in their infants. This high incidence was attributed to an environmental change and Rashid et al. (4) speculated that vitamin D deficiency is less likely to develop in women residing in Pakistan due to the abundant sunshine available. Ahmed et al., in a report on hypocalcemic vitamin D deficiency rickets in breast fed infants in Karachi found severe vitamin D deficiency in all mothers (15/15) in whom serum vitamin D levels were measured (5). As a follow up of this paper the present study was undertaken to find out the vitamin D status of healthy nursing mothers and their infants and factors contributing to any deficiency seen.

Material and methods

From June 1993 to May 1995, serum levels of 25(OH)D were measured in 71 nursing Pakistani mothers and their infants, 6 weeks to 11 months after delivery. Twenty-five and nineteen mothers belonged to the upper and middle socioeconomic classes respectively and were selected from the Well Baby Clinic of the Aga Khan University Hospital, Karachi, Pakistan, where infants are brought in by mothers for immunization after 6 weeks of delivery. Twenty-seven mothers belonged to the lower socioeconomic class and were enrolled from the outpatient of a government general hospital. Forty-eight subjects were included in the study in Summer-Autumn (April to October months) and the rest in Winter-Spring.

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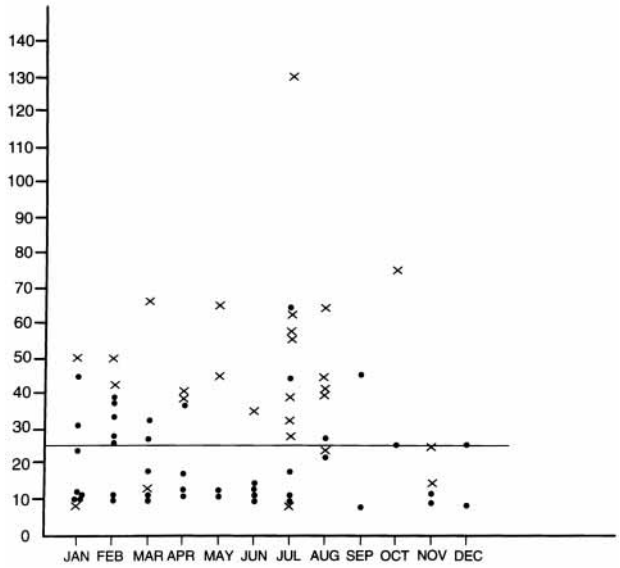

Fig. 1. Seasonal variation of Serum 25 (OH)D (nmol/L) in Mothers of Upper (•) and Lower (X) Socio-economic Class.

A detailed history, level of education of the mother, parity, housing, exposure to sunlight, a twenty-four hours dietary recall and supplemental vitamins if prescribed were noted. Only those mothers who were healthy and had no clinical evidence of vitamin D deficiency were included in the study. The mothers of upper and middle social classes were alike demographically and therefore were merged as upper socioeconomic class (USE). Forty-four (62%) mothers lived in flats and villas and were considered to belong to the USE whereas the 27 (38%) mothers residing in mud houses with one or two rooms opening into streets or small courtyards belonged to lower socioeconomic (LSE) class. The mean (±s.d.) maternal age was 26.62±4.37 years and 55 (77%) had a parity of three and below. The mothers were of average stature and weight and 50 (70%) were educated above the primary school. None of the mothers received vitamin D supplementation during pregnancy and lactation. The women of both social groups wore loose fully covering dresses leaving only face and hands exposed. The diet consisted of wheat based 'chapati' and vegetable curry. Very little unfortified milk was consumed, mostly as tea and yoghurt. Meat, fish and eggs were consumed more frequently by the USE class. The average daily calcium intake of both social classes analyzed in the study was around 500–600 mg. This coincides with the average content of a Pakistani diet as analyzed by the Department of Nutrition at the Aga Khan University Hospital. The average daily dietary vitamin D intake ranged between 100 to 150 i.u. in the USE, the main source being fortified cooking oil and was negligible in the LSE and they mostly consumed unfortified oils.

Blood samples were collected after informed consent of mothers and stored at −20 degrees celsius. Serum 25 hydroxy-cholecalciferol {25-hydroxy vitamin D, 25-(OH)D} levels were measured on all the nursing mothers and infants by the RIA method, Incstar Corporation, Stillwater, Minnesota, USA. The lowest value in adults recorded by this technology is 0.43 nmol/L and the normal range for 25-OHD in May is 68.5±16.5 nmol/L and 59.5±36.5 nmol/L in February. Serum calcium, phosphorus, alkaline phosphatase and alanine transaminase were analyzed by standard technique (Astra system, Beckman Instruments INC, Galway, Ireland). Serum parathormone levels could not be measured because of cost factor and nonreliability of the available kits. The above tests were performed at the research laboratory of the Department of Pharmacology and the clinical laboratory, the Aga Khan University Hospital. The results were analyzed on the statistical program, EPI-info-version 6.02 CDC/WHO October 1994.

Results

Mothers data

The mean serum 25(OH)D in mothers was 36.7±32.4 nmol/L and 34 (48%) had serum 25(OH)D levels in the osteomalacic range (below 25 nmol/L) (Fig. 1). The mean serum 25(OH)D in mothers was not significantly related to their age

Table I. Social and demographic factors related to maternal serum 25 (OH)D (nmol/L)

S. No.	Factor	Patient Numbers	Serum 25(OH)D mean±SD nmol/L	*p* value
1.	Maternal age			
	≤25 yrs	30	28.2±18.9	0.2
	>25 yrs	41	34.3±23.8	
2.	Maternal Education			
	Educated	50	26.2±18.3	0.01
	Uneducated	21	38±26.8	
3.	Maternal Parity			
	≤3	55	27.7±17.8	0.055
	≥4	16	40.5±29.5	
4.	Residence			
	Flat/Bungalow	44	28.3±18.3	<0.001
	Mud Houses	27	33.4±27.2	
5.	Socioeconomic class			
	Upper	44	27±15.5	<0.001
	Lower	27	40.8±28.2	
6.	Season			
	Summer	48	33.8±24	0.5
	Winter	23	27.4±15.4	

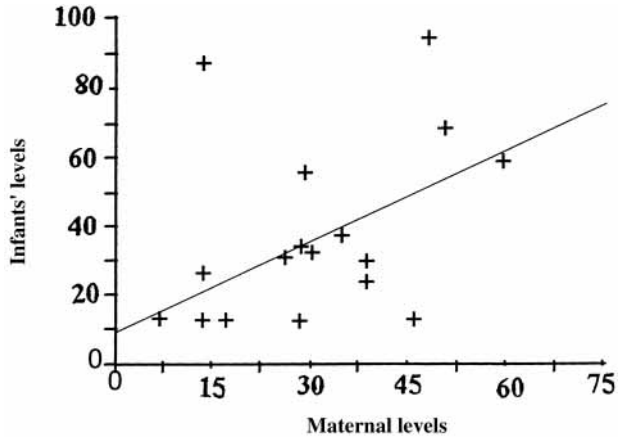

Fig. 2. Correlation between Serum 25 (OH)D levels (nmol/L) of infants below 3 months and their mothers. (r=53%, 95% C.I.=0.15–0.77, p<0.01).

or parity (Table I), but was significantly higher in the uneducated mothers (p=0.01), mothers of lower socioeconomic class (p<0.001) and in those living in mud houses (p=0.001). The mean maternal serum calcium was 2.45±0.19 mmol/L, serum phosphorus was 1.29±0.2 mmol/L and serum alkaline phosphatase was 83.3±84.8 i.u./L.

Infants data

The mean (±s.d.) serum 25(OH)D of infants was 41.25±35.4 nmol/L and 37 (52%) had levels less than 25 nmol/L. Significantly higher values were found in infants examined in Summer (40.7±29.5 nmol/L) as compared to those in Winter (24.46±17.0 nmol/L) (p=0.01). Significant correlation was found between the serum 25(OH)D levels of infants below 3 months of age and their mothers (r=0.53, 95% C.I.=0.15–0.77, p<0.01, Fig. 2).

Discussion

The results of the study confirm the high prevalence of subclinical vitamin D deficiency previously reported in breast-fed infants (5) and healthy and pregnant women (6) in Pakistan. Vitamin D deficiency in nursing mothers has been reported from other tropical countries like Saudi Arabia (7). Although serum 25(OH)D levels were reported to be in the normal range in women residing in Northern Pakistan (4), 35% of these subjects had low vitamin D concentrations (20–45 nmol/L). This study was performed in non-pregnant women and during summer months.

Karachi at sea level and at a latitude of 25°N, has abundant, almost perennial sunshine. However, the preference of the women to avoid direct sunlight, religious and social obligations for the women to cover their skin when going outside their homes, reduces the exposure. In this study we have observed that the amount of sunlight exposure was related to the women's socioeconomic status and therefore the amount of time spent outdoors. The women of USE class who mostly preferred to live indoors, had reduced exposure to direct sunlight and this appears to be the likely explanation for high prevalence of vitamin D deficiency found among them. Significantly higher levels in women of LSE class, among uneducated mothers and those residing in mud houses (both being surrogate of lower social class) can only be explained by increased exposure to sunlight which appears to be related to more time spent outdoors. However, in them the exposure to sunlight was just enough to maintain marginally normal levels of vitamin D.

The diet of our women is deficient in vitamin D and calcium. The mean daily dietary vitamin D intake of Pakistani women was estimated to be 1.05±0.80 μg as compared to 2.67±2.0 μg consumed by British women (4). Pakistani women, in another study, were found to consume 793 mg of calcium daily as compared to 1143 mg consumed by Norwegian women (8). In our study we have found women of USE class to consume only 150 i.u. of vitamin D daily. 'Chapati', a wheat based staple food in Pakistan (a high extraction cereal), has a high content of phytate and inositol hexa and penta phosphate which chelate calcium and diminish its intestinal absorption (4). High fibre vegetable diet has been shown to reduce the serum half life of 25(OH)D, the major circulating metabolite, by approximately one-third (9). Low calcium and high phytate intake would increase the endogenous production of one, 25 dihydroxy cholecalciferol, thereby increasing the metabolic clearance of it precursor 25(OH)D. Deficiency occurs when supply is reduced or the demand increases, as in pregnancy and lactation or during winter months when exposure to sun is minimal.

Parathormone is a sensitive indicator of subclinical vitamin D deficiency (2). We did not measure the serum parathormone level in this study, but in an earlier report from Karachi, and also from Norway, serum parathormone level was shown to be raised in 30% Pakistani women who had low levels of serum vitamin D (5, 9, 12). Forty-eight percent of mothers in this series had very low levels of serum 25(OH)D (less than 25 nmol/L). Raised serum parathormone concentration have been reported in subjects with serum 25(OH)D levels below 37.5 nmol/L (10). In the present series 64% of the mothers had less than 37.5 nmol/L.

The subjects in this study did not receive vitamin D supplementation during pregnancy or lactation. Our finding of a direct correlation between the vit-

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amin D levels of mothers and their infants under 3 months (Fig. 2) indicates that maternal deficiency existed during pregnancy and the infants had low levels at birth. Unless supplemented with vitamin D or exposed to sunlight, these infants are at high risk of developing vitamin D deficiency and rickets (3, 11, 12, 13).

Conclusion

Our finding of subclinical vitamin D deficiency in 48% of nursing Pakistani women and 52% of breastfed infants must be considered an important issue in the maternal and child health care programs, creating awareness among obstetricians, pediatricians and general practitioners. Prevention would require health education regarding sunshine exposure, modification of dietary habits by increasing the calcium intake, and supplementing all pregnant and lactating mothers with vitamin D daily.

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Address for correspondence:

Dr. Mehnaz Atiq
Assistant Professor
Department of Pediatrics
The Aga Khan University
Stadium Road
Karachi-74800
Pakistan

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